

WATER IS LIFE, FOOD IS SURVIVAL

A Study of Everyday Insecurities in South African Households Using Afrobarometer Round 9 Data

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Abstract

This paper investigates the relationship between water insecurity and food insecurity in South Africa using Afrobarometer Round 9 [2022] Survey data. The assessment centres around the question of how water insecurity and other socio-demographic factors influence the likelihood of experiencing food shortages in households, particularly in consideration of South Africa's developmental goals under the National Development Plan 2030. Following conceptual frameworks that view water and food insecurity as interrelated and influenced by socio-demographic factors, the paper's analysis employs a cross-sectional quantitative approach that uses descriptive statistics, bivariate associations and multinomial logistic regression models to examine the statistical associations between them across the [n=1, 542] South African survey respondents. The findings provide evidence that empirically demonstrate the significant association between household level water insecurity and food insecurity in South Africa, as the analysis confirmed that water insecurity is not only statistically associated with food insecurity but it remains a robust predictor even after controlling for socio-demographic factors. The analyses also found that, besides water insecurity, the significant predictors of food insecurity are income insecurity, asset score, higher/post-secondary education, and rural residence. These results suggest that higher income insecurity increases the risk of food insecurity, while higher education and greater asset ownership reduce it, and living in rural areas lowers the likelihood of high food insecurity compared to urban areas.

Declaration

I, Xolani Charles Ncala, hereby declare the contents of this research proposal to be my own work. This Capstone Report is submitted for the degree of Master of Arts in E-Science at the University of Witwatersrand. This work has not been submitted to any other university, or for any other degree.

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Chapter 1

Introduction and Central Research Question

1.1 Introduction

Africa faces a growing water crisis with over 300 million of its people not having access to clean water and more than 700 million without privilege to adequate sanitation (African Union, 2023). The African Union (2023) reported that currently only “\$10 to \$19 billion is invested” each year which is far short of the estimated amount of \$50 billion needed every year to achieve the continent’s water related Sustainable Development Goals (SDGs) This shortfall in the continent’s water funding coupled with “the problem of climate change partly explains growing rate of the water crisis”. The African Union also indicated that “Without action, climate change will make water shortages worse and lead to more food insecurity, disease burden, human displacement and conflict” (African Union, 2023). Given this context, it would be more insightful to examine the specific relationship between these two important social stressors on African individuals and households, specifically how water shortage or insecurity leads to more food insecurity. This provide insights into the everyday impacts of water related challenges to worsening food insecurity.

The country of interest under examination for this study is South Africa. This is mainly because of the fact that South Africa is one of the continent’s most industrialised and largest economies. Considering this, we could expect that it possesses the fiscal and institutional capacity to invest meaningfully in water infrastructure. According to the nation’s Department of Water and Sanitation’s Annual Report (2022), South Africa’s internal development policy framework the National Development Plan 2030 (NDP), outlines specific milestones to achieve its SDG 2030 goals, which include; ensuring that every South African can obtain clean running water within their residences; achieving a food trade surplus, with at least one-third coming from small-scale farmers or families; and guaranteeing food and nutritional security at home (SA National Planning Commission, 2012). The country’s General Household Survey (GHS) (2023) reported that households with water access inside their houses, including households who rely on on-site or off-site water sources, increased by 2.6% (from 84,4 % in 2002 to 87% in 2023) (Statistics South Africa, 2023). However, during this period statistics also showed that water access significantly declined in more than half of the country’s 9-nine provinces. With regards to the process of guaranteeing food and nutritional security for the citizens statistics also indicated that approximately 24% of households across the nation

deemed their food access as “inadequate or severely inadequate” while approximately 76% of households experienced adequate food access (Statistics South Africa, 2023). The country’s planned developmental objectives when viewed with the GHS statistical trends indicate that a slow progress has been made regarding the household’s water access and food security.

Ahmadi (2019) researched on the association between water insecurity and food insecurity among Sub-Saharan African countries, used Afrobarometer survey Round 6 for bivariate and multinomial regression statistical analyses, and concluded that “water insecurity was strongly associated with food insecurity”. The researcher used relatively older data, for this reason, the current study focuses on the most recent period covered by the Afrobarometer Household Survey (AHS), using the South Africa Round 9 survey data from 2022. This contemporary collection of data enabled the micro data analysis, which investigates the association between households’ perspectives on water and food insecurity, to explore how one might affect the other in isolation and when factoring in of socio-economic and demographic variables.

This report begins with the Introduction in Chapter 1-one, which provides the background of the study, its main research questions, and accompanying sub-aims. This followed by Chapter 2-two, the literature review section on the socio-economic impacts of water insecurity and food insecurity, as well as the ways in which they intersect and influence one another. Chapter 3-three covers the proposed methodology and design for the study, which outline the process and procedures to be used for collection and analysis of the Afrobarometer round 9 data. Chapter 4-four covers the analyses section of the paper, which begins with descriptive statistics of the sample, followed by the bivariate analyses table, and then analysing the tables of summary reporting on the results of the Multinomial Logistic Models (MLMs) used to look at the predictive strength and direction the independent variables have on the dependent variable of the study. The last section of the report is Chapter 5, which provides a well rounded summary of the descriptive and statistical analyses conducted, then reports on the implications that the study results have for the South African population and the republic’s developmental aspects.

1.2 Research Questions

1.2.1 Main Research Question

1. What is the association between individual’s perspectives on water insecurity and food insecurity from the Afrobarometer Round 9 South Africa (2022) data?
- With this question the aim is to examine the statistical association and predictive significance between respondent’s perspectives on water insecurity and food insecurity within the broader context of South Africa’s socio-economic landscape and declining water availability using the South African Afrobarometer data sample.

1.2.2 Supplementary Research Question

1. To what extent do households’ perspectives on water insecurity remain significant predictors of households’ perspectives on food insecurity when accounting for

key socio-economic and socio-demographic variables such as gender, location of residence, employment status, income availability and assets?

- With this question the research aims to assess whether household perceptions on the experience of water insecurity remain statistically significant predictors of the experience of food insecurity when after factoring in key socio-economic and demographic factors. This is to determine whether water insecurity has an independent impact on food insecurity, beyond the variability explained by other factors.

Chapter 2

Literature Review: Linking Water Insecurity and Food Insecurity

This literature review explores studies related water and food insecurity and how they intersect to shape the daily realities or struggles of South African households. It highlights the important conceptual underpinnings of studies related to nationally representative data used to examine household experience of both water and food insecurity, their causes, and their socio-economic and health impact on the nation's society. The [section 2.1](#) provides a conceptual exploration into the causes and effects of water insecurity provided by existing water insecurity studies, and the general profile of the water access problem in South Africa. The second part, [section 2.2](#), discusses significant conceptual underpinnings of food insecurity from Southern African and Sub-Saharan Africa studies on factors causing and affecting food insecurity, and the various ways the phenomenon has affected South African individuals and households in detail. The third part [section 2.3](#), of the literature review reports on the views and insights of multiple authors that have previously contributed to the study of the interconnection between water insecurity and food insecurity, their shared socio-economic and socio-demographic drivers, and the impact these dual insecurities have on the individual's everyday experience of essential deprivation. The last part of the the review, [section 2.4](#), provides the key insights and concepts that form the conceptual framework of the study and its further contributions within existing development literature about individuals' experience of food insecurity and water insecurity in the South African context. It also presents the articulated hypotheses that guides the empirical analysis of the study's contribution to understanding the Afrobarometer Round 9 sample respondents' experiential dimensions of food and water insecurity within the Republic.

2.1 Understanding Water Insecurity

A plethora earlier studies has shown that water insecurity is relatively a recent concept in development studies, which means there is still no universally accepted definition of the term. The various existing definitions usually spark debates that contribute to the ongoing evolution of the concept. The conceptualization of water insecurity is often inconsistent and is significantly shaped by contextual and disciplinary differences in how water usage is understood. A commonly referenced definition by Muller et al. (2009) describes water insecurity as lack of “access to sufficient and safe water for

health, livelihoods, ecosystems, and productive uses”, along with “the lack of capacity to manage-water related risks to individuals, the environment and the economy”.

This sentiment was also expressed by Sinyolo et al. (2014) who undertook to explore water insecurity among rural households in South Africa’s uMzinyathi District. The research focused on assessing the factors influencing water insecurity within irrigation schemes and how the variations in water security affect the livelihood of farming households (Sinyolo et al., 2014). The study surveyed a randomly selected sample of 185 households participating in irrigation activities within the Tugela Ferry Irrigation Scheme. Principal Component Analysis (PCA) and the Ordinary Least Squares regression (OLS) were used to analyse the survey data, the findings indicated that the several factors such as “age of farmer, income from non farming sources, membership in farmers association” and others were significantly associated with lower levels of water insecurity (Sinyolo et al., 2014). The study’s findings indicate that households who reported to experience severe water insecurity were likely to have their consistent supply of water needed for farm hold irrigation disrupted, leading to reduced agricultural productivity due to failure of crops (Sinyolo et al., 2014). This means that households have to implement alternative means to provide irrigation for their crops, in instances where there is little or no water.

Against this definition of Muller et al. (2009), Sinyolo et al. (2014) criticise its national-level orientation and how it considers a mixture of macro-scale biological elements and social elements affected by water insecurity. They argue that it overlooks the micro-scale localised realities of water access, particularly for smallholder farmers possibly fall part of irrigation schemes. This implies that although an integrative approach to water security is crucial, a more “narrow approach seems necessary to conceptualise water insecurity” to fit their study of a rural context (Muller et al., 2009; Sinyolo et al., 2014). Therefore, they adopt a definition of water insecurity defined as the “access by the irrigating households to sufficient and reliable water to meet their agricultural needs and their ability to assert their water rights against other parties” (Sinyolo et al., 2014). This definition is especially relevant to their study which focused on rural smallholder farms, primarily because rural areas predominantly struggle with access to water due to the lack of or limited water infrastructure, resulting in the increased reliance on irrigation. However, the research involved a national sample, which provides a narrow applicability to rural households and struggles to suit a broader demographic and geographic context.

On the other hand, recent research indicates that compared to other types of material-need insecurity, the understanding and measurement of water insecurity is more advanced. Various literature contributions have continued to make progress by clearly defining and creating ways to measure it, at both household and individual levels. Frongillo (2023) provides findings on the societal impacts of water insecurity on the health and well-being of households across different contexts. For this author water insecurity includes key aspects like access, availability, usage and stability. The identification of key aspects also signifies the move from the historical evaluations of water insecurity from being focused on physical indicators such as regional water supply, proper sanitation, and community level water access, to recent standardized tools that effectively measure how individuals and households personally experience water insecurity across different cultural and geographic settings (Frongillo, 2023). Currently, there exists standardised tools like experience-based measures to help us to document how individual households perceive, experience and respond to water insecurity, in a quan-

titative manner. Bulled (2017)'s article "The Effects of Water Insecurity and Emotional Distress on Civic Action for Improved Water Infrastructure", a medical anthropology paper, indicates that these approaches originate from the use of "ethnographic evidence on water insecurity...to develop locally grounded scales of water insecurity". The anthropologists and social impact experts construct these scales from the ground level, often starting with qualitative investigations into water usage within specific cultural settings and then they end up having "experience based measures of water insecurity" (Bulled, 2017). According to Frongillo (2023) and Bulled (2017) these methods reveal that water insecurity should be understood as a multidimensional phenomenon with far reaching consequences, which undermine nutritional adequacy, compromise physical and mental health, reduce economic productivity and increase political vulnerability of the population. It is evident that there is a progressive emphasis of the examination of water insecurity through a social lens, as most scholars and researchers find it "useful for informing and evaluating interventions to improve water supplies" (Bulled, 2017; Frongillo, 2023; Sinyolo et al., 2014).

Multiple causes have also been identified for the problem of water insecurity, as it results from the interplay of environmental conditions, engineered infrastructure, and governance systems. Frongillo (2023) indicates that it is natural features like rainfall patterns and land topography form part of the environmental context, while the man-made structures such as canals and reservoirs constitute the physical infrastructure. This distinction is important because it advances the point that many national water policies have failed because they concentrate predominantly on technical and physical solutions while they neglect the social and institutional dynamics that affect water security (Frongillo, 2023). There are various factors that hinder and affect people's ability to have access to a constant and sustainable supply of water, for instance: (a) "the set of attributes associated with the water source itself for instance its economic accessibility and operational reliability"; (b) "the socio-economic factors such as the household's financial capacity to secure safe drinking water"; and (c) "households without piped water connections often pay more for their water than those with a connection" (Ngarava et al., 2019). These factors are mentioned in "Gendered Water Insecurity: A Structural Equation Approach for Female Headed Households in South Africa", where the authors mention that (d) gender factors into the experience of water insecurity, arguing that "women are disproportionately impacted by resource scarcity, yet global indicators of water access often fail to capture the gender-specific burdens and benefits embedded in everyday water-related experiences" (Ngarava et al., 2019). Collectively, these interrelated factors reveal how environmental, infrastructural, socio-economic, and gendered dimensions converge to shape unequal experiences of water insecurity across individuals and societies.

South Africa is part of the ranks among the world's most driest countries faced with significant challenges in water availability and distribution, and its population continues to face problems with regards to the the historically uneven access to basic water services (Ngarava et al., 2019). South African residents are estimated to consume approximately "25 litres of water per person daily", according to Nguvara Ngarava et al. (2019) if these consumption patterns persist, we should expect that the country is expected to face a shortfall of around 17% in available water resources by 2030. It is significant to mention South Africa's complex socio-political history of systematic racism and gender inequalities continues to influence the current water governance policies and social realities related to water supply and access. These historical injustices have

contributed to disputes over land and water access, and continue to hinder progress towards equitable distribution of water across gendered, racial, and economic class groupings. In this instance the principle of equity is still one of the most stressed when it comes to the nation's water policy framework (Ngarava et al., 2019). That is why several researchers advocate for a multidimensional approach that integrates social realities, institutional reforms and infrastructure development to enhance water security (Frongillo, 2023; Muller et al., 2009). In line with this perspective, this study investigates the ways in which water insecurity and its related socio-demographic factors possibly affect the food insecurity of South African respondents from the Afrobarometer Round 9 Survey data sample.

2.2 Understanding Food Insecurity

The conceptual framework required for the examination of food insecurity in Southern Africa is properly set out in a book by Y.Abdalla (2007), which primarily focuses on assessing the causes of food insecurity within Southern Africa and explores the effective policy interventions usually implemented to address the issue. This book employs a qualitative literature review approach that depends on the analysis of document data from various publications to conduct a descriptive and analytical investigation of the problem. The author's general findings reveal that food insecurity in the region has been largely driven by historical land policies, inadequate agricultural growth, poverty, socio-economic challenges, and natural disasters, therefore indicating their importance especially with regards to efforts at improving food availability (Abdalla, 2007). Abdalla (2007) initially shows the importance of analysing food insecurity across two different levels;

- the macro-economic level where food security is understood as the consistent availability of food sufficient to meet nutritional needs of a population and;
- the micro-economic level where food security is contingent upon three related conditions: adequate national food availability, stability in supply and the regular access to food by all households and individuals.

This division proves to be significant because it differentiates that food security at the national level does not automatically translate into food security at the household level. With that said, there is also a second division between two types of food insecurity; "the chronic food insecurity", which is characterized by persistent food insecurity often rooted in systematic poverty and socio-structural inequalities; and the "transitory food insecurity" which arises from short term shocks such as droughts, income fluctuations, or pricing anomalies, which temporarily limit food access (Abdalla, 2007). This set of distinctions highlights which of the two types of food insecurity will more likely require long-term poverty alleviation interventions or short-term emergency interventions.

Abdalla (2007) goes on to define food security as "access by all people at all times to enough food for an active and healthy life". This definition is also widely adopted by the World Bank policy on food security since it considers the essential element of "the availability of food and the ability acquire it" (Abdalla, 2007; Reutlinger & Pellekaan, 1986). It aligns with the Food and Agriculture Organisation's (FAO, 1996) definition of food security, as the physical and economic access to sufficient foods that meets dietary needs and cultural preferences. An important consideration should also be given to the three core components of food security, being income, access and utilisation. Ab-

dalla (2007) emphasizes that practically, without sufficient income and reliable access to water, food security cannot be achieved. Usually in contexts where these resources are limited individuals face increased risk of experiencing food insecurity. With that said, another consideration is that food security must not be solely equated with either agricultural development or agricultural output, this is because it just only accounts for output, which does not automatically guarantee food access for people in society (Abdalla, 2007). In this sense the literature critiques the simplistic intervention approaches that focus solely on increasing agricultural output without addressing the social, economic and political factors that mediate food access (Frongillo, 2023). This shows that the concept of food security has changed significantly since its initial formulation, it not only moved from focusing on objective indicators to subjective perception, but also that "Food security is no longer seen simply as a failure of agriculture to produce sufficient food at the national level, but instead a failure of livelihoods to guarantee access to sufficient food at the household level" (Devereux and Maxwell, 2001 cited in; (Abdalla, 2007)).

Continuing with this, in a recent scholarly work, Walsh and Van Rooyen (2015) also align their understanding of food insecurity with the FAO, for them, the availability, affordability, and accessibility of food within the households constitutes food security. This study on household food security and hunger in the Free State province of South Africa, focused on assessing the rates of household hunger, food insecurity, and their impact in relation to sociodemographic status (Walsh & Van Rooyen, 2015). Individuals aged between 25 and 64 were selected using a cluster sampling technique based on proportional representation, then surveys were administered to collect data on demographic characteristics, food insecurity and hunger, while data on child hunger was compiled through the Community Childhood Hunger Identification Project (CCHIP). The research found that a considerable number of households experienced food shortages, with 47% in rural areas and notably a higher 81% in the urban areas of the province, which indicates greater food insecurity among urban households (Walsh & Van Rooyen, 2015). This similarly confirms Abdalla (2007)'s idea that food insecurity results in severe hunger and malnutrition. Walsh and Van Rooyen (2015) also link to Abdalla (2007) considerations of the factors that affect food security, indicating that "on a global scale these include population growth, climate change, and environmental degradation".

Different definitions of poverty make it difficult to grasp the scope and causes of poverty in South Africa and by extension food insecurity. According to Walsh and Van Rooyen (2015) historically, rural populations were primarily linked to food poverty, however because of growing urbanisation, there has been a discernible change with rising rates of poverty in the country's urban regions. Although access to agriculture such as crop farming does not guarantee to eradicate poverty, the results from support Walsh and Van Rooyen (2015) argument that rural populations with access to agriculture and farming resources are less likely to suffer food insecurity. With that said, Walsh and Van Rooyen (2015) study offers comprehensive knowledge to ways that food insecurity presents itself in rural farm-households. The literature indicates that poverty and food insecurity in South Africa has been a developmental problem that country has been fighting to eradicate since the beginning of the constitutional democratic regime after 1994.

As mentioned in section 2.1 related to water insecurity, South Africa's legacy of Apartheid racialised inequality continues to shape patterns of poverty and access to re-

sources, majority of the previously disadvantaged groups are still faced with persistent structural barriers and growing poverty (Sekhampu, 2013). Indicative of this has been the prioritisation of solving the problem food insecurity through constitutional provisions and legislative policy, as "Food security is part of the section 27 Constitutional rights in South Africa" (Sekhampu, 2013). According to Sekhampu (2013) "Determination of Factors Affecting The Food Security Status of Households in Bophelong South Africa", national level statistics shows that the country is able to "adequately produce main staple foods and possesses the capacity to import additional supplies if needed", however "household level analysis reveals a different picture". From this Sekhampu (2013) was able to propose that "it is necessary to investigate how the resources of a household determine its access to food", in order to better understand the different levels of food insecurity South Africans as a whole experience. In the analyses of his work, he conducted survey interview questions for the township of Bophelong, and later used the data to formulate a Binary Logistic regression model "to determine the effects of some socio-economic and demographic characteristics of the households on their food security status" since the outcome variable is dichotomous between those who are food insecure and those who are food secure (Sekhampu, 2013). The results of the model revealed that household food insecurity is significantly influenced by income of the household, its size, the age of the breadwinner or head, employment gender and the marital status of the household head. Among these the most the variable with the most effect was the household income, where higher income is positively linked to better food security outcomes. These are the similar conclusions which this paper comes too but the most significant variable of interest, not mentioned in Sekhampu's study is water insecurity and the effects that it has on food insecurity when considering the effect of socio-economic and demographic factors (Sekhampu, 2013).

2.3 Intersection of Water Insecurity and Food Insecurity

Unlike when considered separately, there are a few studies that investigate the intersection and association between household water and food insecurity. The conceptual framework to advance the amount of research related to this intersection is outlined in "Concurrence of water and food insecurities, 25 low-middle income countries" (Young et al., 2022). The study aimed to investigate "how water insecurity and food insecurity are associated within their nationally represented samples" to characterise how much these two issues co-occur, and to investigate whether those people who are water insecure are more likely to be food insecure (Young et al., 2022). The researchers of this study made use of data collected from the World Gallup Poll (WGP) of 2020, which contained answers from [n= 31,755] respondents who completed both the Individual Water Insecurity Experiences Scale and the Food Insecurity Experience Scale. The statistical analyses conducted, used linear regression models and multivariate logistic models, which found that approximately 66.8% of the respondents who reported water insecurity, 18.3% of them also reported food insecurity (Young et al., 2022). These findings show a very significant association between water insecurity and food insecurity proving that inadequate access to clean water negatively influences people's food intakes. According to Young et al. (2022), the lack of access to water can potentially disrupt essential household functions for instance cooking a meal, which often forces the households to rely on other alternatives that do not require water to prepare or

skip the meal entirely. They also highlight that environmental issues like floods and severe droughts may also cause difficulties for households to produce food for market or subsistence (Walsh & Van Rooyen, 2015; Young et al., 2022). The financial impact is also particularly noteworthy, since the increase in water costs may result in less money for food, while labour-intensive water collection procedures may restrict time spent on earning money (Young et al., 2022).

However, moving on along this line of enquiry, Ahmadi's (2019) analysis of the association between food insecurity affected by water insecurity and other socio-demographic factors in 32 sub-Saharan African countries, takes a micro-level household centric approach through the analysis of survey data. His focus is on areas that face major obstacles in both water and food access, because they help understand how water and other relevant factors, affect households' ability to obtain enough food to meet their nutritional needs (Ahmadi, 2019; FAO, 1996). The main dependent variable to the author's study was a proxy question from the survey to measure how many times individuals and their families have gone without food in the past one year, with response values "0= Never" to "5= At least once a month" (Ahmadi, 2019). The main predictors included a proxy question to measure water insecurity and income availability both ranging from "0= Never" to "5= At least once a month", while the controls were sex, age, employment, area of residence, and asset ownership. Since outcome variable for Ahmadi (2019) differed from the conventional binary responses by having degrees of the experience of food insecurity, one of the main methods of analysis besides frequency analyses and bivariate associations, include running two Multinomial Logit Models (MLMs) which produce odds ratios to determine whether the one unit change in the values of water insecurity is associated with a change in the odds of the predictor variable food insecurity. The results revealed that there is a strong significant association between water insecurity and food insecurity, as it showed that respondents who were water secure were 34 $OR = 34.08$ times more likely to experience no form of food insecurity (Ahmadi, 2019). The socio-demographic factors were found to also have substantial role in contributing to food security, particularly income availability which has a positive effect in being food secure, while individuals without any formal education had a likelihood of experiencing food insecurity at $OR = 0.141$ and those residing in rural areas with $OR = 0.457$ (Ahmadi, 2019). This current study treads on more similar steps of statistical enquiry with Ahmadi (2019) who used data from Round 6 of the Afrobarometer Household Survey 2016, but its analyses are particularly examine the significant impact of water insecurity on food insecurity for the sample of selected from the democratic Republic of South Africa round 9 survey.

While the link between water and food may appear simple, it is usually materially shaped by complex environmental and socio-economic factors. For instance, water is fundamental to food security because agriculture depends heavily on quantity and available quality of water. This means that water plays a crucial role in food security, especially in places where many people rely on farming subsistence food production (Abdalla, 2007; Ahmadi, 2019). This relates to fact that the environmental factors like climate change and drought, affect the available amount of water. Frongillo (2023) advances this argument and highlights the intersection between food security and water security in three ways, mainly; "(1) they are closely intertwined and can manifest simultaneously within the same households or individuals; (2) they coexist and function as compounding stressors that negatively affect well-being; (3) they are linked through

various physical, biological and behavioural pathways (unsafe water may lead to illness like cholera, which disrupts individual's nutrition)".

2.4 Study Contribution and Analytical Focus

Despite the growing academic focus on food and water insecurity, a key issue that has been noted is that there is less development literature that focus on how these two crises intersect to impact the daily living of households than there is literature on their separate aspects, which might be troubling considering the high level of theoretical association that can be drawn between two. This means that most of the shortfalls might be related to the measures that will be sufficient to use when building on their empirical relationship. Important to note is that both food and water insecurity studies have now moved from focusing only on the "objective indicators"(standardised measurements/tools) to also integrating the focus on the "subjective perception"(personal judgement/experience) of households (Abdalla, 2007).

Most studies as seen from the Ahmadi (2019) and Young et al. (2022)), offer important continental or cross-national perspectives on food and water insecurity, which have been significantly presented generalized finding across diverse African regions, that will prove to be useful for my study which will consider South Africa's unique socio-political and socio-demographic realities affecting the double insecurity problem. Although Walsh and Van Rooyen (2015) provide valuable insights to the country's internal food insecurity and household vulnerability, they rarely treat water insecurity as a primary variable of interest or explore how water deprivation might exacerbate the nation's food insecurity in tangible ways. As a result, the relationship between water and food insecurity is frequently addressed in parallel rather than in a truly intersectional and integrated manner.

The research in this paper contributes to the advancement of the broader topic and knowledge by examining the interconnection between water insecurity and food insecurity at the household level in South Africa, using nationally representative sample data from Afrobarometer Round 9. It positions itself at the intersection of micro-level water and food insecurity, thereby potentially contributing to bridging their theoretical and empirical elements in literature and social statistics. As highlighted by all the authors reviewed in this section it is important to consider controlling for socio-demographic factors and comparing households with differing water access systems (internal vs external) or income availability, because it will provide a more nuanced and contextually grounded reality of this disparity Abdalla (2007), Walsh and Van Rooyen (2015), and Young et al. (2022). The main hypotheses of the paper were formulated based on the research questions and the literature review to provide guidance to the statistical analyses that in section 4.1:

- **H0 Null Hypotheses** : None of the included variable predictors (e.g, water insecurity, income insecurity, education level, asset score) are significantly associated with food insecurity.
- **H1: Alternative Hypotheses** : At least one of the predictors is significantly associated with food insecurity

The contributions and results of this report not only enriches academic enquiry but it also equips policymakers with evidence-based insights to potentially design interven-

tions that respond to the dual crises of food and water insecurity as interconnected realities facing millions of South African households today.

Chapter 3

Proposed Methodology and Research Design

3.1 Research Design

This study employs a cross-sectional quantitative research design to investigate the relationship between household-level water insecurity and food insecurity in South Africa, using the Afrobarometer Round 9 survey (2021–2023), a nationally representative dataset. According to Jepson et al. (2021) the cross-sectional approach is appropriate as it allows for the analysis of data collected at a single point in time, thereby enabling the identification of correlations between variables without inferring causality. This design provides a foundational understanding necessary for exploring how these intersecting vulnerabilities manifest within South African respondents as they exist at the single moment recorded between 2021 and 2023 (Felton, 2024).

3.2 Data Source and Sampling

The study relies on quantitative data from Afrobarometer Round 9 Felton (2024), which by the organisation's procedures, allowed to be collected through a stratified multistage sampling method to ensure national representativeness (Ramah-Attoh & Robertson, 2014). Approximately [n = 1, 580] South African households were selected randomly within stratified enumeration areas defined by province, urban-rural status, and socio-economic background. Data collection included face-to-face interviews in multiple local languages, with standardized questionnaires addressing water access, food security, and socio-economic conditions. As observed in Ahmadi (2019) study, the rigorous sampling and data collection process of Afrobarometer ensures that diverse household experiences are recorded, which allows the generalisation to the population, increasing relevance and reliability of findings. The fact that Afrobarometer (Felton, 2024) survey data has multiple variables including key aspects that are potentially associated with food insecurity and water insecurity eliminated the work of having to use more than one data set to form the response variable and predicted variables needed for the analyses of the paper.

3.3 Variables and Data Analysis

3.3.1 The Dependent Variable and the Main Independent Variable

The statistical analyses focuses on food insecurity as the dependent variable and water insecurity as the primary independent variable, alongside key socio-demographic controls including gender, location of residence, employment status, and income availability. Food insecurity is measured by frequency of insufficient food access, while water insecurity is seen as inconsistent access to safe and sufficient water (Ahmadi, 2019). In this study food insecurity is measured using Afrobarometer question [Q6A]: “In the past 12 months, how often have you or anyone in your household gone without enough food to eat?” with response options ranging from 0=Never, 1=Just once or twice, 2=Several Times, 3=Many times, 4=Always (Felton, 2024). On the side of the predictors water insecurity is measured by question [Q6B]: “How often in the past 12 months have you or anyone in your household gone without enough water for household use?” with similar frequency response options (Felton, 2024).

When considering the idea that the number of participants in the Round 9 survey are less than 10 000, it makes it a small sample which might have not be easy to generally interpret when it is spread across multiple categories that are not strictly ordinal, and it will also decrease the analytical value of both the response and the independent variables. The remedy to improve the analytical potential of the variables and the potential of sparse cell counts across multiple categories (Ahmadi, 2019), the variables were recoded into a three-level outcomes. Food Insecurity was pooled as; 0= No food Insecurity taking in the values of Never; 1= Low Food Insecurity taking in the values of Just once or twice and Several times; 2= High food insecurity taking in the values of Many times and Always. Water Insecurity was also similarly pooled as; 0= No Water Insecurity taking in the values of Never; 1= Low Water Insecurity taking in the values of Just once or twice and Several times; 2= High Water insecurity taking in the values of Many times and Always.

3.3.2 The Socio-economic and Demographic Controls

Most studies of food insecurity as seen in [section 2.2](#) have found that it does not occur isolated from other social, political and geographical phenomena, as these have the possibility to impact the different levels of food insecurity an individual faces. Most form of statistical analysis which are popular among food insecurity always consider the effect of such factors and possibly apply them as controls (Young et al., 2022). For this the study some of control variables include income insecurity, employment, water source location (e.g, internal vs. external) and others below, (see, [Table 3.1](#)) all of which are extracted from the the same Afrobarometer survey dataset (Felton, 2024).

Table 3.1: The Table of Socio-economic and Socio-Demographic Predictors

Predictor	Description
Gender	Binary indicator of the respondent's sex (Male or Female).
Urban/Rural Residence	Binary indicator of geographic region (Rural or Urban).
Education Level	Highest level of formal education completed, from no schooling to tertiary.
Employment Status	Current labour force status (employed, unemployed).
Income Insecurity	Whether the respondent reports going without a cash income in the past year. (No Income security, Low Income Security, High Income security)
Primary Water Source	The location of household water source (Inside compound or outside compound).
Age	Respondent's age in years.
Asset Score	Composite index based on ownership of 4 household assets. (Radio, Television, Motor Vehicle and Mobile cellphone)

3.3.3 Statistical Analyses

The main objective of the study as set out in [section 1.2](#), is to investigate the relationship between household water insecurity and food insecurity levels in South Africa in the context of the country's declining water availability. For this examination to be employed to capture the exact numbers or proportions of the interest variables, check for existing bivariate associations and potential multivariate effects. The statistical computing environment used for the statistical analyses is R.studio (R.4.4.2) which includes an interpreter for the R programming language that was specifically designed for statistical computing. According to Fox and Weisberg (2018) it is a "powerful environment that simplifies the exploration of data by making it possible for users to easily view and interact with data tables, complex graphical visualisations and summary statistics", it therefore maintains a good standard for advancing statistical interpretation and data driven decision making.

Statistical analyses for most survey data usually begin with a crucial step of assessing the frequency distribution of the variables that are used in the study, to ensure that there are no data anomalies, and to understand the variable distributions. This is also the starting point of the statistical analyses for this paper. The frequency distributions are displayed in [section 4.1](#) below. The frequencies generally tell us how much of the respondents reported having experienced low food insecurity, no food insecurity or high food insecurity, a simple univariate exploration, worth reporting on, especially for primary variables of interest.

The second part of the data analyses is particularly focused on testing the bivariate associations between the dependent variable food insecurity, the independent variable water insecurity, and the other categorical predictors. In a paper by Duke et al. (2020), they explain the Chi-square as the inferential statistics used to test the association between two categorical variables, "to help decide whether sampled differences across

categories can be attributed to random chance or are likely due to genuine difference”. In this investigation the use of the chi-square test allowed us to infer from the sample of respondents whether the differences in food insecurity status are related to the differences in the water insecurity within South Africa’s population. The strength of such relationships between categorical variables is usually assessed through the phi test and Cramer’s V, the only difference is that “phis is used with 2x2 contingency tables while Cramer’s V is applied to larger tables” with more than 4 variables (Duke et al., 2020). Their similarity is that they both provide standardised measure which takes on any values between 0 and 1, with the moderate association starting at 0.3 and strong association starting at 0.5 (Duke et al., 2020). The results of this test guided the selection of the categorical variables that were included in the Multinomial Logit Model (MLM) used to examine the predictive significance of the predictors on the dependent variable.

The multinomial MLM regression analysis forms the final part of the statistical analyses performed on the sample data, from the nature of the variables of interest in the study being “nominal unordered categories” and only two numeric predictors (Kwak & Clayton-Matthews, 2002). In their article titled “Multinomial Logistic Regression” Kwak and Clayton-Matthews (2002) explain that when the data exhibits “characteristics of categorical variables which are truly discrete, nominal, and unordered” the MLM becomes the appropriate analytical approach. Generally multinomial models are an extension of the binomial glm model where predictor has binary outcomes (e.g, yes/no), if outcomes are more than two categories have, there is a selected reference category - by analytical convenience - that allow for the estimated model coefficients to be interpreted relative to it (Duke et al., 2020). For this study the categories that are taken as reference for the independent and dependent variables are “No water insecurity” and “No food insecurity” respectively see section 4.1.

The R function for running a multinomial model found from the nnet package by Ripley et al. (2016) is multinom(), which can “fit models in which observations represent counts in the several response categories, as one would have in a contingency table, or individual responses” (Fox & Weisberg, 2018). The output summary of the model lays out the log-odds coefficient for each independent variables along with its p-value which indicates the statistical significance of the association. According to Kwak and Clayton-Matthews (2002) the best practice of interpretation for such output is to exponentiate them which transforms their scale into odds ratios, for example when interpreting the coefficient for age in one category of the response variables the coefficient must be exponentiated to get odds ratios, preferably known as the Relative Risk Ratio (RRR) scale for MLMs. The RRR basically reflects the observed change in odds of being in a selected category compared to the reference group for each unit increase in the predictor variable (Kwak & Clayton-Matthews, 2002). Taking the advice of Fox and Weisberg (2018) it is usually challenging to estimate the effects of multinomial model variables just through regression summaries and coefficients, so the regression for the reduced rather than the full model in table() will be visualised on a MLM effects plot Figure 4.2 on section 4.3.

The formula below best represents the multinomial logit model in equation format, where the left hand side of the equation represents the log-odds for the probability of being in the response level m compared to being in the reference level (Fox & Weisberg, 2018). On the right side of the formula is a linear equation of the intercept $\beta_0^{(m)}$ and the predictor variables $X_1, X_2, \dots, X_k$. The important point of focus is on the resultant

coefficients β which inform us how much a variable changes the odds of being in one category compared to the reference on the outcome variable (Fox & Weisberg, 2018).

$$\log \left(\frac{P(Y = m)}{P(Y = \text{ref})} \right) = \beta_0^{(m)} + \beta_1^{(m)} X_1 + \beta_2^{(m)} X_2 + \cdots + \beta_k^{(m)} X_k$$

3.4 Ethical Considerations and Study Limitations

The Afrobarometer Round 9 dataset is publicly available and anonymized, ensuring that no personal identifiers are linked to the survey responses to minimize the potential harm of the sample respondents (Felton, 2024; Ramah-Attoh & Robertson, 2014). In the instances that this paper is primarily a secondary data analyses it qualified for an ethical clearance waiver, the researcher has taken the necessary steps to comply to Afrobarometer’s data usage terms and conditions. Data protection and confidentiality are strictly maintained throughout the analyses process. Additionally, the analyses respects the principles of academic integrity and transparency as outlined in the North-West University ”Policy on Academic Integrity” (North-West University, 2024). A potential limitation of this study, is the reliance on self-reported data, which may result in social desirability bias (Ahmadi, 2019; d’Agostino et al., 2025; Young et al., 2022). However, the robust sampling strategy and the representative sample size for a cross-sectional analysis selected from the Afrobarometer data enhanced the validity and generalizability of the findings to the South African population (Felton, 2024).

Chapter 4

Statistical Analyses Results

4.1 Basic Descriptive Statistics

This is the start of the statistical analyses for the Afrobarometer South Africa Round 9 data, where the frequency distributions of the multiple categorical variables are reported. This is a preliminary analyses of seeing and just reporting on the actual numbers of people distributed across the various categories of the variables of interest and the controls. As we can see from [Table 4.1](#) out of the expected data sample of $n = 1,580$ respondents the Total Sample Size examined for this analyses is slightly reduced to $n = 1542$ after coding away the non-response variables where some people chose not to answer or respond to certain questions of the survey.

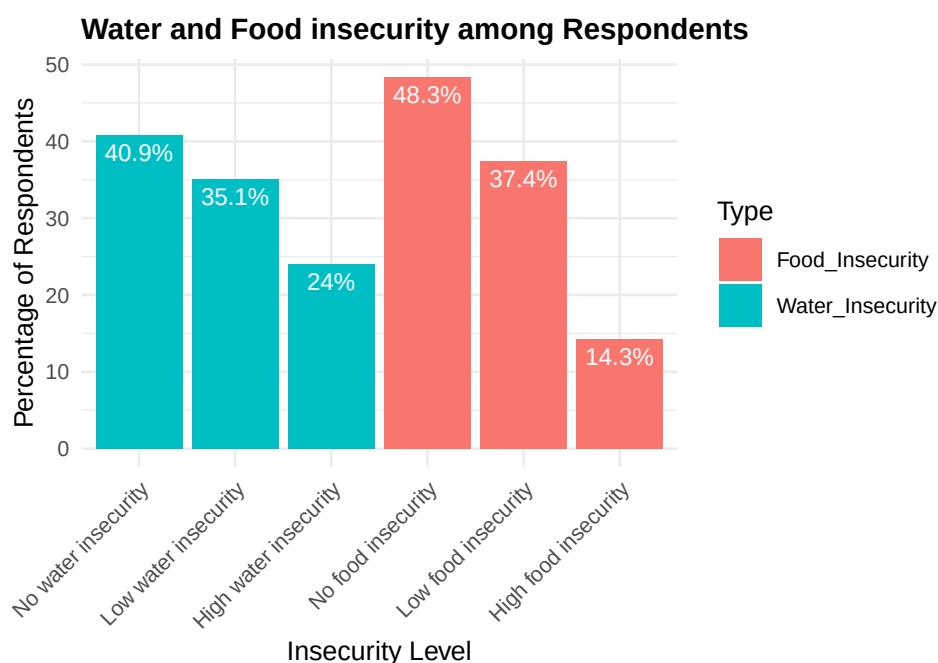


Figure 4.1: Respondent Water and Food Insecurity Percentages in South Africa.
Source: Afrobarometer Round 9, South Africa 2022

In terms of the prevalence of water insecurity the study's main independent variable the numbers shows that out of the total sample size, approximately 40.9% of the respondents reported having no experience of water insecurity (see, [Figure 4.1](#)), while

for the low water insecurity category it was about 35.1% of the respondents, and the remaining 24% of the respondents fell in the high water insecurity rate (see, [Figure 4.1](#)). These sample counts are not good in terms of the experiences of not having water in South Africa because this means that close to 59% of households reported experiencing some form of water insecurity between the years 2021-2023. Moving to the main dependent variable, food insecurity on the right panel of [Figure 4.1](#), it shows that approximately 48.3% of the respondents were food secure, while for those who experienced some form of food insecurity the distribution reports 37.4% of respondents experienced low food insecurity and the rest of the 14.3% experienced high food insecurity. On the side of food insecurity things seem almost balanced as a little more than half of the respondents report having experienced some form of food insecurity over the past year.

In terms of the numeric predictors, the average age of the respondents was 40 years, with a spread of about ± 16 years away from the mean, meaning that most of the respondents fall between the mid 20s and mid 50s (see, [Table 4.1](#)). For the Primary Asset score between 0 to 4 the average for the variables is close to 3 with a standard deviation of ± 1 , meaning that most people own between 2 and 4 of the primary assets.

Moving to the categorical control variables in [Table 4.1](#) we can see that most of the respondents originally fall on the urban geography at 68% and only 32% ≈ 493 are sampled from rural areas, while on the water source variable 82.5% of the respondents had a water source within the compound they lived in and leaving the remaining 17.5% with external water sources (see, [Table 4.1](#)). This is one of the interesting finds because 82.5% for respondents with an internal water source from the Afrobarometer survey sample (see, [Table 4.1](#)), is close to the 87% of people with water access inside their houses reported by the General Household Survey in 2023 (Statistics South Africa, 2023). When looking at one of the socio-economic indicators, 34% of respondents reported having no income insecurity (never gone without cash income), while most of the distribution experienced low income security at 39% (going without cash), and the rest 26% experienced the highest level of going without cash income. Education level shows that about 26.4% have gone through higher post-secondary education, while only 49 respondents (3.2%) had never received formal education, and most of the distribution (70.4%) has some primary or secondary level education. There is not much difference between the gender category and the sample of the respondents had roughly the same equal numbers of man and woman.

4.2 Categorical Bivariate Associations

The [Table 4.2](#) presents the statistical the results of testing for bivariate associations between the dependent variable food insecurity and some of the categorical predictors. As mentioned in the methods [subsection 3.3.3](#), the Cramer's V is a statistical estimate for a chi-square tests of association between two categorical predictors, it serves for inferential purposes to help decide whether differences across categories are by random chance or likely due to genuine difference, and takes on any values between 0 and 1. From the results (see, [Table 4.2](#) income insecurity has the strongest association observed with food insecurity with Cramer's $V = 0.381$, which confirms some of the convictions in the literature that the group of people who experience income insecurity (without cash) are more likely to report food insecurity at $p < 0.01$, showing that the

Table 4.1: South Africa's Sample Characteristics of Respondents from Afrobarometer Round 9 Survey

Variable	N (%) or Mean $\pm$ SD	Category
Age	39.9 $\pm$ 15.7	
P.Asset_Score	3.22 $\pm$ 0.9	
Food_Insecurity	745 (48.3%)	No food insecurity
	577 (37.4%)	Low food insecurity
	220 (14.3%)	High food insecurity
Water_Insecurity	630 (40.9%)	No water insecurity
	542 (35.1%)	Low water insecurity
	370 (24%)	High water insecurity
Income_Insecurity	531 (34.4%)	No income insecurity
	608 (39.4%)	Low income insecurity
	403 (26.1%)	High income insecurity
Urban_Rural	1049 (68%)	Urban
	493 (32%)	Rural
Education_Level	49 (3.2%)	No formal schooling
	1086 (70.4%)	Primary/Secondary education
	407 (26.4%)	Higher/Post-sec education
Gender	770 (49.9%)	Man
	772 (50.1%)	Woman
Employment_Status	980 (63.6%)	Unemployed
	156 (10.1%)	Employed part time
	406 (26.3%)	Employed full time
Primary_Water_Source	1272 (82.5%)	Internal
	270 (17.5%)	External
Total Sample Size	1542	

Reference: Afrobarometer Round 9, South Africa 2022.

observed association is not due to random variation. The main independent variable water insecurity has the second strongest association with food insecurity with Cramer's $V = 0.318$, which in practice means that for the South African sample, households that experience going without water are more likely to experience food insecurity at $p < 0.01$. With this finding, we confirm one of the main questions of the article which assess whether the experience of not having water has a potential association with the experience of respondents going without food, a finding that aligns with Ahmadi (2019)' study.

Table 4.2: Bivariate associations between food insecurity and socio-demographic factors in South Africa ($n = 1,542$). Strength measured using Cramér's V for categorical variables (Chi-square tests) and Pearson's r for continuous variables.

Predictor	Association	P-value	Strength	Significance
Categorical predictors (Cramér's V)				
Water Insecurity	0.318	0.0000	Moderate	Significant
Gender	0.028	0.5470	Negligible	Not Significant
Urban/Rural Residence	0.070	0.0225	Negligible	Significant
Education Level	0.233	0.0000	Moderate	Significant
Employment Status	0.148	0.0000	Weak	Significant
Income Insecurity	0.381	0.0000	Moderate	Significant
Primary Water Source	0.157	0.0000	Weak	Significant

Reference: Afrobarometer Round 9, South Africa 2022.

The level of education shows up as having a moderate significant association to food insecurity with a lower Cramer's V of 0.233, which suggests that there is a possibility that lower levels of education are linked to the high levels of food insecurity within the sample. The rationale for this is set by Young et al. (2022) that as individuals progress to higher levels of education, it increases the chances of employment and income. The results for the Gender variable show that it does not have a significant effect on the level of food insecurity for this sample with Cramer's V close to 0 and $p = 0.547$. The effect size for the urban-rural divide variable is also close to 0 but has a statistically significant $p = 0.0225$, although its effect is close to negligible it is still an effect which can be considered with some significance. The reason why we can still consider geographic location to have some effect on respondents' food insecurity is because there might be other factors that mediate this relationship, the multinomial regression models in the next section will help us determine this conviction and also give a proper response to the hypotheses mentioned in [section 2.4](#).

The check for the bivariate analyses between the outcome of food insecurity and the numeric predictor variables Age and Asset score also necessitated the formulation of an MLM, because the test for association between them means running separate multinomial regression for the two, respectively. In this case, they are just applied as controls, so it is best to examine their association in the multinomial regression with all the other variables of interest in the full model on [section 4.3](#), rather than testing for their association separately. This is one of the main reasons why there two MLMs on the next [section 4.3](#), just so we can test the associations of the predictor variables to food insecurity in a controlled model [Table 4.3](#), and then look at the practical effects of

only the predictors found to be significant determinants of household food insecurity [Table 4.4](#)

4.3 The Multinomial Logistic Models (MLMs)

4.3.1 Full MLM vs Parsimonious (Reduced) MLM

Table 4.3: Adjusted Multinomial Logistic Regression Estimates

Predictor	Low food insecurity	High food insecurity
(Intercept)	0.37 (0.0452) *	0.14 (0.00232) **
Water Insecurity: Low	3.24 (3.11e-15) ***	3 (9.48e-06) ***
Water Insecurity: High	3.08 (9.02e-10) ***	7.86 (4.44e-16) ***
Ref: No Water Insecurity		
Residence: Rural	0.74 (0.0476) *	0.51 (0.00158) **
Ref: Urban		
Employment: Part-time	0.9 (0.615)	1.13 (0.676)
Employment: Full-time	0.89 (0.497)	0.82 (0.432)
Ref: Unemployed		
Income Insecurity: Low	4.03 (< 2e-16) ***	2.01 (0.00851) **
Income Insecurity: High	6.72 (< 2e-16) ***	14.27 (< 2e-16) ***
Ref: No Income Security		
Age (Continuous)	1 (0.912)	1 (0.806)
Water Source: External	1.26 (0.204)	1.34 (0.232)
Education: School	1.2 (0.637)	0.94 (0.892)
Education: Higher/Post-secondary	0.44 (0.0436) *	0.18 (0.00152) **
Ref: No Formal Education		
Asset Score (Continuous)	0.77 (0.000978) ***	0.76 (0.00768) **
<i>Model Statistics: LogLik = -1193.76, AIC = 2439.52, BIC = 2578.38, McFadden's R^2 = 0.224</i>		

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Reference: Afrobarometer Round 9, South Africa 2022.

On this part of the report the focus is on the two Multinomial Logistic Models formulated to test the following hypothesis:

- **H0 Null Hypotheses** : None of the included variable predictors (e.g, water insecurity, income insecurity, education level, asset score) are significantly associated with food insecurity.
- **H1: Alternative Hypotheses** : Atleast one of the predictors is significantly associated with food insecurity

With reference back to [subsection 3.3.3](#) of the methodology, the multinomial regression is suitable because it can fit models where the outcomes represent counts in several categories, and then the output communicates the log-odds coefficient for each independent variable. In this case it tells us the chances of being in in one category of food insecurity whether "Low food insecurity" or "High food insecurity" compared to the reference base which "No food insecurity", all reported in log-odds which can be exponentiated to odds ratio or RRR. The [Table 4.3](#) above provides the regression summary estimates in an already exponentiated relative risk ratio scale. The basic results show

that we can reject the **H0 Null Hypotheses** and accept the **H1: Alternative Hypotheses**, since there is a presence of more than seven statistically significant variables on both levels of food security compared to the reference.

As the bivariate analyses has proven Water Insecurity, Income Insecurity, Education Level and geographical area are amongst the most significant predictors of food insecurity in the model with very low p values. There is also new insight offered about the association of the dependent variable to the numeric predictors Age and Asset score, and it seems that only Asset Score has predictive significance of food insecurity. Without the need to interpret the coefficients of the model we can already tell from the presence of an asterisk (*) next to the p-value inside the parentheses whether a variable and its different levels is regarded as a significant predictor. From the results of the full model, some of the variables are insignificant predictors of food insecurity including Employment status, continuous variable Age, water source and the primary/secondary school education Level with p-values $p > 0.05$. The next step is to run a parsimonious MLM for this data, which means that Table 4.4 of the regression excludes the insignificant predictors to help improve the explanatory power of the model while reducing the random noise of insignificant predictors. Since one of the levels of education has predictive significance, it is not excluded, and the education variable itself helps add on to the analytical significance of the results, particularly since the results show that having gone through some Post-secondary/higher education has a significant affect on both the low and high food insecurity level.

Table 4.4: Reduced Multinomial Logistic Regression Estimates for Food Insecurity in South Africa

Predictor	Low food insecurity	High food insecurity
(Intercept)	0.4 (0.0365) *	0.16 (0.00148) **
Water Insecurity: Low	3.26 (1.55e-15) ***	3.08 (5.38e-06) ***
Water Insecurity: High	3.17 (2.34e-10) ***	8.24 (0) ***
Ref: No Water Insecurity		
Residence: Rural	0.79 (0.0927)	0.54 (0.00226) **
Ref: Urban		
Income Insecurity: Low	4.06 ($< 2e-16$) ***	2.04 (0.00709) **
Income Insecurity: High	6.88 ($< 2e-16$) ***	14.61 ($< 2e-16$) ***
Ref: No Income Insecurity		
Education: Primary/Secondary	1.14 (0.72)	0.89 (0.797)
Education: Higher/Post-secondary	0.4 (0.0227) *	0.15 (0.000509) ***
Ref: No Formal Education		
Asset Score (Continuous)	0.76 (0.000473) ***	0.75 (0.00411) **
<i>Model Statistics: LogLik = -1195.66, AIC = 2427.32, BIC = 2523.46, McFadden's $R^2 = 0.222$</i>		

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Reference: Afrobarometer Round 9, South Africa 2022.

The more parsimonious reduced model is displayed in Table 4.4, taking a first glance at which of the predictors does not retain an asterisk (*) that represents the significance level of the association with the dependent variable, we can see that for a respondents who falls on the category of rural, the variable has lost the significant predictive association of experiencing low food insecurity, and only retained its predictive significance for the category of High food insecurity relative to no food insecurity. The education

variable does not change the form and order of its significance as it only retains it in the higher education level. The effect of dropping the insignificant predictors from the full model has not affected much of the model estimates and statistics of fit. The model statistics shows us that in mathematical terms the full model versus the reduced, do not have much difference between them, for instance the Log-likelihood measure of how well the model explains the response variables does not differ for [Table 4.3](#) (-1193.74), and [Table 4.4](#) (-1195.66). The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) which usually balances the model fit and complexity of the MLMs, is lower and on the reduced model [Table 4.4](#) and therefore shows it has a better fit, because it achieves similar explanatory power with fewer parameters. As a measure of better fit the McFadden pseudo R^2 within each model shows us how much of a better fit is the model compared to the null model. For [Table 4.3](#) the full model, it is standing at $R^2 = 0.224$ this indicates it is a marginally better fit compared to the null model, but because it includes some insignificant predictor variables like Age and Employment status its explanatory power has potential to advanced by excluding them. For [Table 4.4](#) the parsimonious model, the McFadden R^2 is at 0.222 which closely similar to the full model, but because it has less variables it remains a statistically better model that is easier to interpret the significant factors affect food insecurity levels of households.

4.3.2 The Relative Risk Ratio (RRR) and Predictor Effects

Overall the model statistics show that [Table 4.4](#) is a better model to help us interpret the most significant factors or predictors that influence respondents' experience of food insecurity. With this regard, it is the best model for which we can interpret and visually show the relative risk ratio (RRR) of the predictors and their direction of effect towards the dependent variable, see [Figure 4.2](#) below.

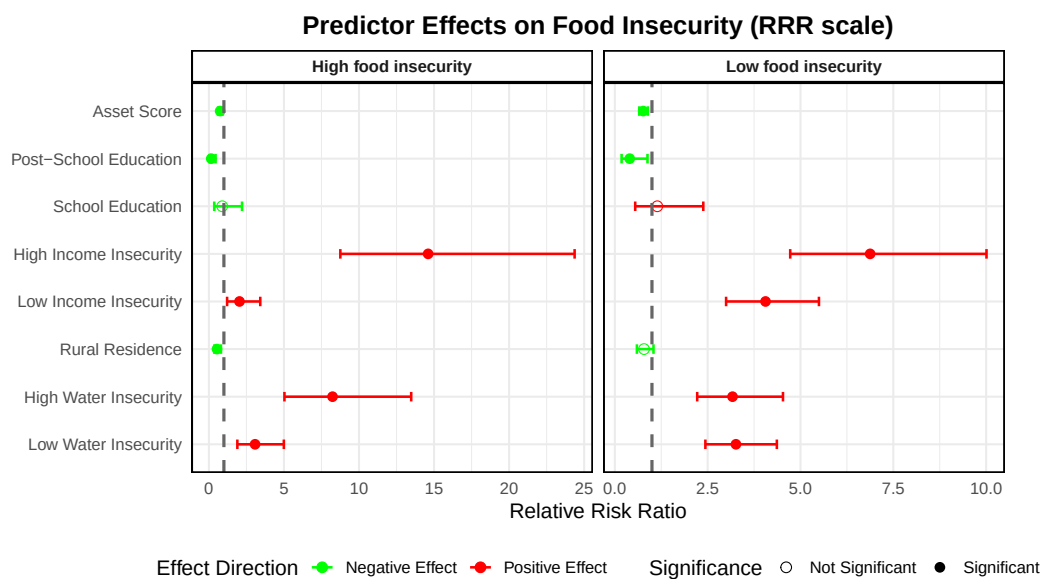


Figure 4.2: Predictor Effects on Food Insecurity (Reduced Model)

For the effect of the main independent variable water insecurity, on the different levels of food insecurity, the regression summary in [Table 4.4](#), shows us that it when it is low it has a $RRR = 3.26$, which means that individuals experiencing low water insecurity are ≈ 3 times more likely to experience low food insecurity than those with no water insecurity at a highly significant effect $p < 0.001$, and for those respondents who report experiencing high water insecurity the $RRR = 3.17$, indicating that they are ≈ 3 times more likely to experience low food insecurity compared relative to those with no form of food insecurity also highly significant with $p < 0.001$. This relationship is well represented by [Figure 4.2](#), we can see that on both the food insecurity levels low and high water insecurity have a significant positive effect on increasing the experience of food insecurity. The output presented in [Figure 4.2](#) shows that respondents who fall under the category of high income security have the higher RRR on both the levels of food insecurity, this means that being on this category significantly puts respondents on high risk of experiencing food insecurity relative to no food insecurity, for instance respondents who experience high income insecurity have an RRR of 14.61 at $p < 0.001$ meaning that individuals within that category are 14 times more likely to experience high food insecurity relative to individuals without any water insecurity - a highly statistical significant association. This then shows that water and income insecurity on all levels have significant positive contribution on respondents' experience of food insecurity.

When moving to the variables that visually have a negative or green effect on food insecurity we can see that Post-school education has the highest negative effect on the experience of food insecurity, because on the reduce model the RRR for this variable is ≈ 0.4 with $p < 0.05$, we can say that practically belonging to the category of those who pursued higher/post-secondary education you are $\approx 60\%$ less likely to experience low food insecurity and $\approx 85\%$ less likely to experience high food insecurity at a highly significant $p < 0.001$ compared to those without any formal education. This shows that with higher levels of education respondents are less likely to experience both forms of food insecurity. The primary/secondary level of education relative to no formal education shows to not have any meaningful effect on both levels of food insecurity, visual representation shows that it has a non significant positive effect on low food insecurity relative to no food insecurity, while it has a negative non significant effect on the high food insecurity level.

The primary asset score also has a negative effect on respondents' experience of low food insecurity reporting a $RRR = 0.76$ at $p < 0.001$, while reporting a $RRR = 0.75$ on the experience of high food insecurity, practically a one unit additional move on asset score means that respondents are 24% less likely to experience low food insecurity relative to no food insecurity, and $\approx 25\%$ less likely to experience high food insecurity. Finally when looking at the geographic region variable we can see that being rural areas of South Africa has a non significant negative effect on respondent's experience of low food insecurity with $RRR = 0.79$ with $p > 0.05$, while being in the rural areas shows to have a negative significant impact on respondents' experience of high food insecurity with a $RRR = 0.54$ at $p < 0.001$, which practically means that rural respondents compared to urban respondents are $\approx 46\%$ less likely to experience high food insecurity relative to no food insecurity. This result slightly confirms the predisposition that in the South African sample the urban areas have high numbers of food insecure individuals than the rural geographies, due to the high population counts within them. Overall the model has shown some of the potentially significant contributors to both the high

food insecurity levels and the low food insecurity levels, and the results are satisfactory which allow us to make similar inferences about factors affecting food insecurity in South Africa at the population level from using the Afrobarometer sample data. [chapter 5](#) reconciles the statistical analyses with the broader theoretical convictions outlined in this paper regarding water and food insecurity in South Africa, offering an interpretation of whether the empirical results align with the theoretical framework and other statistical summaries related to the topic.

Chapter 5

Conclusion

South Africa has seen slow progress towards some of its objective milestones set out in the NDP 2030, which include ensuring that every citizen can receive clean running water within their residences, achieving a food and trade surplus, and guaranteeing food and nutritional security (SA National Planning Commission, 2012). The country's general statistics indicate that from the year 2002 to the year 2023, the number of households with water access in their houses only rose 2.6% from 84.4% to 87% , while approximately 24% of households across the nation deemed their food access as severely inadequate (Statistics South Africa, 2023). Against this backdrop the main purpose of this data study, was set to investigate the statistical association between the household level water insecurity and food insecurity, by using the nationally representative data from Afro barometer Round 9 survey (Felton, 2024). This was to determine whether water insecurity remains a significant predictor of food insecurity even after accounting for key socio-demographic and socio-economic factors.

The progression of the report analyses from the descriptive statistics (section 4.1, the bivariate association analysis (section 4.2, and to the two multinomial models presented in section 4.3, has help achieve the aim to empirically demonstrate the significant association between household level water insecurity and food insecurity in South Africa, as the analysis confirmed that water insecurity is not only statistically associated with food insecurity but it remains a robust predictor even after controlling for socio-demographic factors. In this sense the results and findings of the study generally align with existing literature, that has found a significant association between household experience of food and water insecurity (see Table 4.4). In practical terms we can imagine that respondents or households that are often faced with inadequate access to water often struggle to prepare meals that require boiling or washing, therefore leading to them to skip meals or rely on less nutritious alternatives. The Food and Agricultural Organisation (FAO), suggest that with regards to food production, the limited water access tends to undermine crucial subsistence farming and home gardening initiatives therefore increasing household's dependency on external food sources (FAO, 1996).

The visual description of the reduced model results as shown by Figure 4.2, indicate that although water insecurity has a positive effect on food insecurity on relative risk ration scale, the variable with the highest significant effect on food insecurity is income insecurity. This reflects the direct impact of going without cash on the household food access, in the sense that when individuals or families have an unreliable source of income, they are likely to have struggles with having an adequate supply of food needed. Since South Africa is well known for its high unemployment which means that

the majority of the population does not have a stable income, such a result highlights the importance of government institutions such as the South African Social Security Agency who are responsible for administering and distributing social assistance grants that help alleviate poverty and economic strife from society. With regards to the effect of education on food insecurity, the analyses of the MLMs show that respondents with post-secondary education are significantly less likely to experience food insecurity, this effect might be attributed to the widely available fact that higher education has possibility to improve people's employment and income prospects. In the South African setting, tertiary studies are still inaccessible to many due to the expensive fees needed to cover for the period of study, as a result it is seen as inaccessible to many, and therefore, the active promotion for free higher education for citizens may also be taken as a protective intervention to prevent poverty and in effect food insecurity.

Although most conversations around food insecurity and water insecurity would say that by being in a geographical locations such as the rural areas puts you at a disadvantage of experiencing more of these phenomena than the urban counterpart, this particular study on the round 9 South Africa sample revealed that rural respondents are generally less likely to experience food insecurity than the respondents that occupy the urban areas. This might be attributed to the fact that urban areas are usually densely populated, have higher food prices and rates of poverty, and there is less space or land for a kind of subsistence living. Such results may explain and necessitate the existence of food based organisation in urban areas such as the Johannesburg Food Bank that administer and give out food parcels to the financially needy and homeless persons. Finally, the findings showed that the primary asset score has a consistent negative effect on food insecurity, this implies that having material assets such as a mobile phone or a car can serve as buffers against the experience of deprivation as they generally improve access to information and services. This finding practically supports the idea of asset-based poverty alleviation strategies, to help uplift individuals beyond just income alone. A general example of such strategies perhaps include the provision of bicycles to rural school going children, just so they can avoid the troubles of not going to school or convenient stores because of long distances.

When looking at the significant findings reported on this paper one can see that at the South African Afrobarometer survey data sample can enable us to examine one of the most important associations between multiple variables, that can be recoded, modelled, and re-analysed, in this case being food insecurity, water insecurity, income insecurity and others. Further research along this line of topic or enquiry one can include an addition of multiple other countries as case studies rather than focusing on one country, in this way the modelling and statistical analyses are not only generalisable to only South Africa one of the most wealthiest nations in Africa. Such problems such as food insecurity and water insecurity particularly need to be examined in countries where they are the actual lived realities of the majority of the society, where the findings will have a greater impact significance on the development of the society.

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Chapter 6

Attached Document- AI Declaration

Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

- | | |
|---|-------------------------------------|
| 1. Idea Generation (research problem/design, hypothesis) | ☒ |
| 2. Sourcing Related Work (summarising, identifying sources) | ☐ |
| 3. Methods and Experiment Design (experiment setup, model tuning) | ☐ |
| 4. Data Analysis (presentation, coding, interpretation) | ☐ |
| 5. Theoretical Development (theorem proving, conceptual analysis) | ☒ |
| 6. Code Development (generating algorithms, writing scripts) | ☐ |
| 7. Presentation (rendering graphics, formatting) | ☐ |
| 8. Editing (grammar, readability) | ☒ |
| 9. Writing (text generation, document structuring) | ☐ |
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If other uses were involved, please specify below:

Generative AI tool used (list all)	Used for?

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number: 2 4 5 6 5 4 3

Candidate signature: X.C Ncala

Date: 21 September 2025